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FORMALIZING THE ISOLATION LEMMA IN ISABELLE/HOL

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ABSTRACT

Randomized algorithms are a cornerstone of modern theoretical computer science [1], yet their proofs often rely on informal probabilistic intuition that obscures critical assumptions about independence and measurability. This report presents a formalization and machine-checked proof of the Isolation Lemma, a key tool in randomized algorithms that isolates a unique minimum-weight solution by randomly assigning weights to elements in combinatorial problems such as the Perfect Matching problem [2], [3]. By carrying out this development in the Isabelle/HOL interactive theorem prover [4], [5], we make all probabilistic assumptions explicit and machine-checkable, thereby strengthening the rigor of the underlying algorithmic reasoning.

We begin by formally specifying the Isolation Lemma, including the underlying product probability space for random weight assignments, the precise statement of the lemma, and the combinatorial constructions required to reason about minimizers over finite sets. Building on this foundation, we develop a complete machine-checked proof by synthesizing and reconciling arguments scattered across multiple informal and folklore-style presentations of the lemma and resolving implicit assumptions present in their informal treatments. Finally, we apply the formalized Isolation Lemma within a two-stage machine-checked correctness argument for a randomized parallel algorithm for the Perfect Matching problem: it first establishes a conditional correctness result for extracting the unique minimum-weight perfect matching, and then uses the formalized Isolation Lemma to justify the required uniqueness premise with high probability, thereby illustrating how such randomized arguments can be integrated into a rigorous formal verification framework.

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I am also grateful for the LAMR reading group on Logic and Mechanized Reasoning, which helped me build background in computational logic, mechanized reasoning, and formal proof methods that proved useful for this project.

My thanks also go to the Isabelle/HOL and Archive of Formal Proofs communities for providing the libraries, documentation, and prior formal developments that made this project possible. Their work has created a rich and reusable formal ecosystem on which this report could build.

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1 INTRODUCTION

Randomization is a central paradigm in modern theoretical computer science (TCS) [1], and many randomized algorithms rely on probabilistic lemmas that enforce useful structural properties under random perturbation. One important example is the Isolation Lemma, which shows that a random weight assignment yields, with high probability, a unique minimum-weight solution within a family of sets.

Such arguments are often presented at a pen-and-paper level, with assumptions about probability spaces, independence, measurability, and minimization left implicit. Formalization provides a way to reconstruct these arguments in a setting where all relevant definitions and proof obligations must be stated explicitly and verified mechanically.

This report studies the formal verification of the Isolation Lemma in Isabelle/HOL and its use in a machine-checked correctness argument for a randomized parallel algorithm for the Perfect Matching problem. The main contribution is not the underlying algorithmic idea itself, but its mechanized reconstruction: first, a conditional correctness argument showing that a unique minimum-weight perfect matching can be recovered by algebraic tests, and second, a probabilistic justification of that uniqueness premise via the formalized Isolation Lemma.

1.1 BACKGROUND

1.1.1 RANDOMIZED ALGORITHMS IN TCS

Randomized algorithms form a major class of algorithms in theoretical computer science. They are often used to break symmetry, simplify constructions, or obtain guarantees that are difficult to achieve deterministically [2], [6].

In the present report, the relevant randomized technique is random weighting: random weights are assigned to basic elements so that, with high probability, a unique minimum solution is isolated among many candidates. This principle is captured by the Isolation Lemma, first introduced by Mulmuley, Vazirani, and Vazirani [2], whose

classical statement and proof idea are reviewed in Chapter 2.

1.1.2 Formalization and Theorem Proving

Formalized mathematics and interactive theorem proving (ITP) provide a framework in which mathematical definitions, statements, and proofs are encoded within a precise logical system and checked mechanically [4]. In this setting, assumptions that are often left implicit in pen-and-paper proofs must be stated explicitly.

This is particularly relevant for probabilistic arguments such as the Isolation Lemma, where correctness depends not only on combinatorial reasoning but also on explicit probabilistic constructions and side conditions. In the present report, formalization is used not as an end in itself, but as a way to reconstruct a classical randomized argument in a fully precise and machine-checked form.

1.2 Project Objectives

The primary objective of this project is to formalize and prove the Isolation Lemma within the Isabelle/HOL theorem proving system. This includes establishing its probabilistic statement, its associated probability bound, and the precise conditions under which uniqueness holds.

To support this development, the project also formalizes the probabilistic and combinatorial structures on which the lemma depends, including random weight assignments, independence of random variables, and minimization over finite sets.

Specifically, this project aims to:

- formalize random weight functions together with their corresponding product probability spaces, thereby making probabilistic assumptions explicit;
- provide a complete machine-checked proof of the Isolation Lemma, including an explicit upper bound on the probability of non-uniqueness and a precise characterization of failure conditions;
- apply the formalized lemma within a machine-checked correctness argument for

a randomized parallel algorithm for the Perfect Matching problem.

1.3 Scope of Work

The scope of this project is limited to theoretical formalization and proof construction within the Isabelle/HOL framework. In particular, this report covers:

- the mathematical formulation of the Isolation Lemma and its formal expression;
- the probabilistic modeling of random weight assignments and their associated properties;
- the formalization of key combinatorial and probabilistic arguments used in the proof;
- a two-stage machine-checked correctness argument for a randomized parallel algorithm for the **Perfect Matching** problem, consisting of a conditional correctness proof for the edge extraction and a probabilistic correctness guarantee derived from the formalized Isolation Lemma.

This work does not address general derandomization techniques for randomized algorithms, nor does it consider implementation-level concerns or performance optimization. While the Isolation Lemma is classically associated with randomized parallel approaches to the Perfect Matching problem, the scope of this formalization is limited to verifying the logical and probabilistic correctness of the isolation step. It does not include the formalization of complexity classes or proofs of complexity-theoretic upper bounds inside Isabelle/HOL. Moreover, the internal implementation details of the Isabelle/HOL system and its underlying logical foundations are beyond the scope of this report.

1.4 Report Outline

The remainder of this report is organized as follows.

Chapter 2 (Preliminaries) reviews the background material required for this work,

including randomized algorithms, probabilistic reasoning, and basic concepts from interactive theorem proving.

Chapter 3 (Modeling) presents the formal modeling framework used throughout the report. It explains how the combinatorial and probabilistic structures underlying the Isolation Lemma are represented in Isabelle/HOL.

Chapter 4 (Proof) contains the core formalization of the Isolation Lemma. The development is organized using a PP/SP/FP structure, corresponding to pen-and-paper proofs (PP), structured proofs (SP), and fully formal proofs (FP).

Chapter 5 (Perfect Matching) develops a two-stage machine-checked correctness argument for a randomized parallel algorithm for the Perfect Matching problem. It combines a conditional algebraic correctness argument with the probabilistic uniqueness guarantee provided by the formalized Isolation Lemma.

Finally, Chapter 6 (Conclusion) summarizes the main contributions of the report and discusses possible directions for future work.

2 PRELIMINARIES

Before turning to the formal development of the Isolation Lemma, we briefly review the background needed in later chapters. This chapter introduces interactive theorem proving, Isabelle/HOL, and the mathematical and algorithmic background relevant to the Isolation Lemma and its application to the Perfect Matching problem. The material presented here is purely preparatory and does not constitute original contributions.

2.1 INTERACTIVE THEOREM PROVING

Interactive theorem proving (ITP) is an approach to formal reasoning in which mathematical definitions, statements, and proofs are expressed within a precisely specified logical system and verified by a computer. Unlike fully automated theorem provers, interactive systems rely on human guidance to structure proofs, while delegating routine logical inference to the underlying proof engine.

A central characteristic of interactive theorem proving is that all logical assumptions must be made explicit. Properties that are often left implicit in pen-and-paper proofs – such as finiteness, non-emptiness, totality of functions, or measurability of events – must be stated precisely and verified within the formal system. As a result, proofs developed in an interactive theorem prover do not appeal to informal intuition, but instead derive correctness from the soundness of the underlying logic.

Interactive theorem proving has been successfully applied across a wide range of areas in computer science and mathematics, including program verification, compiler correctness, and the formalization of foundational mathematical theories. In the study of algorithms, ITP provides a principled framework for reasoning about complex arguments, particularly when proofs involve subtle dependencies between combinatorial and logical assumptions.

From a methodological perspective, ITP also supports the digitalization of mathematical proofs [7]. Formal developments can be stored as machine-checkable artifacts, reused across projects, and extended collaboratively. This supports large-scale proof developments and emerging forms of human–computer and

human–AI collaboration in formal reasoning.

2.2 Isabelle/HOL System

Isabelle is a generic interactive theorem proving framework that supports formal reasoning in a variety of logical systems. Among its most widely used instantiations is Isabelle/HOL, which is based on classical higher-order logic (HOL) [4]. In this setting, logical formulas are constructed using typed λ-calculus, and reasoning proceeds within a sound and expressive classical logic.

Isabelle/HOL provides a rich environment for formal mathematical development. Its logic supports higher-order functions, inductive definitions, and structured proofs, making it well suited for the formalization of mathematical arguments arising in theoretical computer science. The system is accompanied by an extensive standard library covering areas such as sets, relations, algebraic structures, and real analysis.

Of particular relevance to this report is Isabelle/HOL's support for probabilistic reasoning. The system includes a mature probability theory library that formalizes measure spaces, random variables, and probability distributions within the HOL framework [5]. This allows probabilistic arguments to be expressed using explicit constructions of probability spaces and measurable functions, rather than relying on informal probabilistic intuition.

Taken together, these features make Isabelle/HOL a suitable platform for the formal verification of randomized algorithms and probabilistic lemmas. It provides the logical expressiveness required to model combinatorial structures, as well as the mathematical infrastructure needed to reason rigorously about probability and randomness.

2.3 The Isolation Lemma

The Isolation Lemma, originally introduced by Mulmuley, Vazirani, and Vazirani [2], is a probabilistic principle showing that, with high probability, a unique solution exists among a family of sets. It has found important applications in randomized algorithms,

particularly in problems involving combinatorial search and matching.

Informal Statement. Let U be a finite set (called the *universe*), and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets of U. Each element $x \in U$ is assigned a weight $w(x)$ independently and uniformly at random from $\{1, 2, \ldots, N\}$ for some integer $N \geq 1$. The total weight of a set $S \in \mathscr{F}$ is defined as

$$w(S) = \sum_{x \in S} w(x)$$

The Isolation Lemma states that, with high probability, there is a unique set in $\mathscr{F}$ that minimizes this weight.

Theorem (Isolation Lemma) [2]: *Let U be a finite set with $|U| = n$, and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets. Suppose that each $x \in U$ is assigned an integer weight $w(x)$ drawn independently and uniformly at random from $\{1, 2, \ldots, N\}$. Then the probability that there exists a unique set $A \in \mathscr{F}$ minimizing $w(S)$ is at least $1 - \frac{n}{N}$.*

$$\Pr[\exists! A \in \mathscr{F} \text{ minimizing } w(A)] \geq 1 - \frac{n}{N}$$

Proof Idea. The proof analyzes the probability that *two or more* sets in $\mathscr{F}$ attain the same minimum weight. The key observation is that such a collision can be attributed to a specific element whose randomly assigned weight causes two competing sets to tie. By bounding the probability of this event for each element and applying a union bound over the universe, one obtains the upper bound $\frac{n}{N}$ on the failure probability.

Representative Proof (after Spencer [8]). We include this classical argument as background intuition; its fully formal treatment is developed later in the report.

Let $U = \{e_1, e_2, \ldots, e_n\}$ be a finite set of n elements, and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets of U. Suppose each element $x \in U$ is assigned a random weight $w(x)$ uniformly from $\{1, 2, \ldots, N\}$.

For any $x \in U$, define:

$$\alpha(x) = \min_{\substack{S \in \mathscr{F} \\ x \notin S}} w(S) - \min_{\substack{S \in \mathscr{F} \\ x \in S}} w(S \setminus \{x\})$$

The quantity $\alpha(x)$ is determined by the weights of elements other than x itself, and may therefore be treated as fixed when conditioning on all weights except $w(x)$. Since $w(x)$ is chosen independently and uniformly from $\{1, 2, \ldots, N\}$, the probability that $w(x) = \alpha(x)$ is at most $1/N$.

Now suppose there are two distinct sets $A, B \in \mathscr{F}$ such that both minimize the total weight $w(S) = \sum_{x \in S} w(x)$. Let $x \in A \triangle B$ (i.e., an element in the symmetric difference). Without loss of generality, suppose $x \in A \setminus B$. A direct calculation shows that $\alpha(x) = w(x)$ for such an element x.

Thus, if two distinct minimizers exist, then $w(x) = \alpha(x)$ for some $x \in U$. Since this occurs with probability at most $1/N$ for each x, a union bound over n elements yields a total failure probability of at most n/N.

Therefore, with probability at least $1 - \frac{n}{N}$, the minimum-weight set in $\mathscr{F}$ is unique.

Remarks.

- The probability bound $1 - \frac{n}{N}$ improves as N increases.
- The lemma is non-constructive: it guarantees uniqueness with high probability but does not identify the minimizing set.
- The Isolation Lemma is a central tool in randomized algorithms, for example in parallel approaches to the Perfect Matching problem.

This argument is presented at a high level and leaves several assumptions implicit, including the existence of the relevant minima, the independence structure of the weight assignments, and the measurability of the events involved. Making these assumptions explicit is a central goal of the formal development in later chapters.

2.4 Parallel Algorithm for Perfect Matching

2.4.1 The Perfect Matching Problem

Let $G = (V, E)$ be an undirected graph. A *matching* in G is a set of edges such that no two edges share a common endpoint. A matching is called a *perfect matching* if every vertex in V is incident to exactly one edge in the matching. Figure 2.1 illustrates a small bipartite graph together with one perfect matching.

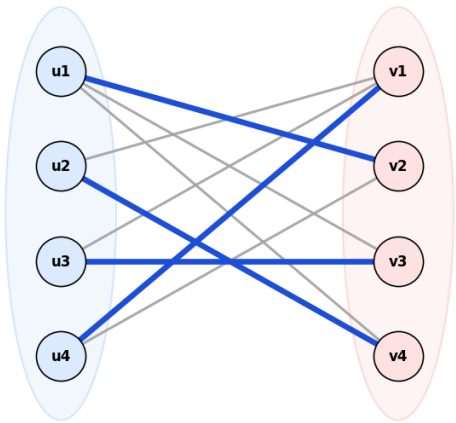

Fig. 2.1: A bipartite graph together with a highlighted perfect matching. A perfect matching selects a set of pairwise disjoint edges that covers every vertex exactly once.

The *Perfect Matching problem* asks whether a given graph admits a perfect matching, and in the constructive setting, how to identify one. This problem occupies a central position in graph algorithms and combinatorial optimization, and has been studied extensively from both complexity-theoretic and algorithmic perspectives [3].

From a combinatorial viewpoint, the difficulty of the problem lies in the global nature of the constraint: a perfect matching must cover all vertices simultaneously, and local choices of edges may interact in complex ways. As a consequence, large graphs may admit many distinct perfect matchings, or none at all, and multiple solutions introduce significant structural symmetry.

In this report, the Perfect Matching problem serves as the main application context for the Isolation Lemma developed in later chapters.

2.4.2 Challenges in Parallel and Algebraic Approaches

One of the main difficulties in the Perfect Matching problem arises from the high degree of symmetry inherent in its solution space. A graph may admit many distinct perfect matchings, none of which is distinguished *a priori*. From an algorithmic perspective, this symmetry complicates both decision and construction, as no local criterion suffices to single out a particular solution.

This issue is particularly pronounced in the context of parallel computation. Parallel algorithms aim to exploit independence and concurrency, but symmetry among solutions often prevents local decisions from composing into a globally consistent matching. In the absence of a canonical choice, parallel processes may conflict or fail to converge on a single valid solution.

From a complexity-theoretic viewpoint, this symmetry barrier is a central obstacle [2], [9] in the design of efficient parallel algorithms for matching. While the Perfect Matching problem is solvable in polynomial time by classical sequential algorithms, highly parallel algorithms require additional structure to break this symmetry in a controlled way.

For this reason, symmetry-breaking mechanisms play an important role in randomized parallel approaches to matching. A classical example is random weighting, which distinguishes one solution from a large family of combinatorially equivalent candidates.

2.4.3 Random Weighting and the Isolation Principle

A common strategy for breaking symmetry in combinatorial optimization is to introduce randomness through *random weighting*. Instead of reasoning about all feasible solutions equally, one assigns random weights to the basic elements of the structure and compares solutions by their total weight.

In the context of the Perfect Matching problem, this means assigning random weights to edges and considering the total weight of a matching as the sum of the weights of its constituent edges. While many perfect matchings may exist in the original unweighted

graph, random perturbation alters the comparison criterion among them.

The central idea behind this approach is that, under a random weighting, ties between distinct solutions become unlikely. Intuitively, two different matchings differ on at least one edge, and the random weight assigned to that edge can be expected to break the tie between their total weights. As a result, with high probability, one matching becomes strictly lighter than all others.

This intuition is formalized by the *isolation principle*. Rather than guaranteeing the existence of a solution, isolation guarantees that, among all feasible solutions, a unique minimum-weight solution emerges with high probability after random perturbation. Crucially, this uniqueness is not imposed deterministically, but arises as a probabilistic property of the random weight assignment.

From an algorithmic perspective, isolation transforms the structure of the problem. Instead of reasoning about a symmetric family of solutions, an algorithm can focus on a single distinguished object: the unique minimum-weight solution. This is particularly important in parallel computation, where global coordination among many equivalent solutions is difficult to achieve.

The Isolation Lemma provides a precise probabilistic statement of this principle. It quantifies the probability that random weighting isolates a unique minimum-weight solution, and identifies the parameters controlling the failure probability.

2.4.4 Role of the Isolation Lemma in Perfect Matching Algorithms

In randomized and parallel algorithms for the Perfect Matching problem, the Isolation Lemma plays a structural rather than a computational role. When multiple minimum-weight perfect matchings coexist, algorithmic tests intended to identify a single solution may become ambiguous, especially in a parallel setting where local decisions must remain globally consistent.

The Isolation Lemma resolves this difficulty by guaranteeing, with high probability, a unique minimum-weight perfect matching after random perturbation. This allows the

algorithm to treat one matching as a distinguished object, thereby removing ambiguity caused by symmetry.

Accordingly, in such algorithms the role of isolation is to provide a distinguished minimum-weight solution with high probability, thereby enabling subsequent algorithmic reasoning.

2.5 Related Work

2.5.1 Related Results in Theoretical Computer Science

The Isolation Lemma belongs to a broader collection of structural tools in theoretical computer science that use probabilistic or algebraic reasoning to derive algorithmic consequences. Such results typically do not solve a computational problem directly; instead, they establish a high-probability or algebraically certified property that enables subsequent algorithmic reasoning.

A canonical example is the Schwartz–Zippel lemma, which bounds the probability that a nonzero polynomial evaluates to zero under a random assignment, and thereby underlies randomized polynomial identity testing [10], [11]. Another important example is Alon's Combinatorial Nullstellensatz, which provides an algebraic framework for certifying the existence of combinatorial objects through nonvanishing conditions on suitable polynomials [12]. In a different direction, Chernoff-type concentration bounds play a similarly foundational role in randomized algorithm design by turning independence assumptions into quantitative guarantees on deviation probabilities [1], [13].

Related themes also appear in hardness amplification and structural complexity results. For example, Yao's XOR lemmas show how probabilistic transformations can amplify weak hardness into stronger forms [14], [15], while results on unique solutions and isolation, such as the Valiant–Vazirani theorem and subsequent refinements, further illustrate the importance of uniqueness as a structural resource in complexity theory and randomized computation [16], [17]. Likewise, spectral and isoperimetric inequalities, such as Cheeger-type bounds and related expander results,

provide another classical instance in which mathematically rich structural statements have important consequences across theoretical computer science [18], [19], [20].

From this perspective, the Isolation Lemma is naturally viewed not merely as a technical fact about weighted set systems, but as one of a broader class of foundational tools that turn randomness into useful structure. Its role in randomized parallel algorithms for the Perfect Matching problem is one particularly influential manifestation of this general theme [2], [3]. In the present report, however, the focus is not on extending the algorithmic applications of the lemma, but on formalizing the lemma itself and making its probabilistic reasoning machine-checkable.

2.5.2 Formalization of TCS and Randomized Algorithms

The present work fits into a broader landscape of formalized mathematics and mechanized reasoning in theoretical computer science. Major proof-assistant systems for such developments include Lean, Isabelle/HOL, and Rocq [21], [22], [23]. Lean is widely used for large-scale formalized mathematics; Isabelle/HOL provides a classical higher-order logic environment with strong support for structured proof development; and Rocq (formerly Coq) is based on dependent type theory. Associated large-scale formal libraries and archives include Lean's *mathlib* and the Isabelle Archive of Formal Proofs (AFP) [24], [25].

Within this broader landscape, Isabelle/HOL and the AFP are particularly relevant to the present report. The Isabelle/HOL environment provides a mature framework for higher-order reasoning [4], while its probability library supports formalizations of measure spaces, random variables, and probabilistic constructions [5]. Related AFP developments further show how these probabilistic foundations can be used to formalize randomized algorithms and probabilistic arguments [26]. More generally, the long-term vision of formalized mathematics as a reusable digital body of verified knowledge has been articulated since the QED manifesto [7], and has since been reflected in major verification efforts ranging from compiler correctness to large formal proofs in mathematics [27], [28].

More closely related to this report are recent formal developments showing that proof

assistants can already support substantial randomized and algebraic reasoning in TCS. For example, the AFP entry on the Schwartz–Zippel lemma formalizes a central probabilistic tool for polynomial identity testing and includes a perfect-matching-related application in bipartite graphs [29]. The AFP entry on executable randomized algorithms further demonstrates that randomized procedures can be reasoned about formally while remaining connected to executable artifacts [26]. Other recent AFP developments include formalizations of Rabin's closest-pair algorithm and the Sumcheck Protocol, illustrating that both nontrivial randomized algorithms and probabilistic algebraic proof techniques can be mechanized in Isabelle/HOL [30], [31].

These developments provide important context for the present project. They show that proof assistants can already handle sophisticated probabilistic, algebraic, and algorithmic reasoning, and that Isabelle/HOL in particular provides a sufficiently mature infrastructure for such work. However, to the best of our knowledge, the Isolation Lemma itself has not previously been formalized. The contribution of this report is therefore not to introduce the algorithmic use of isolation in perfect matching, but to mechanize the lemma itself and use it within a machine-checked correctness argument for a randomized parallel algorithm for the Perfect Matching problem.

3 Formal Modeling of the Isolation Lemma

This chapter begins the formal development by translating the core components of the Isolation Lemma into precise definitions in Isabelle/HOL. Building on the informal statement of the lemma provided in Chapter 2, this chapter introduces the precise formal model used in the later proof development. At this stage, the relevant objects are defined explicitly in Isabelle/HOL, while the main correctness results are established only in Chapter 4.

This chapter develops the formal model of the lemma component by component. First, we define the underlying universe of elements and the associated weight functions, and explain the choice of natural numbers for weight representation. Second, we formalize the notion of a minimizer and the condition of its uniqueness within a family of sets. Finally, we construct the product probability space using the HOL-Probability Library, ensuring that the weights are distributed uniformly and independently.

These definitions form the foundation for the correctness proof in Chapter 4. Special attention is given to finiteness assumptions and the construction of measure spaces, since both are central to the later formal proof.

Standing assumptions. Throughout this chapter, we fix a universe U and a non-empty family of sets $F \subseteq \mathscr{P}(U)$, and assume that U is finite. For the randomized construction, we fix an integer $N \geq 1$ and sample weights for all $x \in U$ independently and uniformly from $\{1, \ldots, N\}$.

3.1 Modeling Weights and Universe

Informal descriptions of randomized constructions often begin with intuitive assumptions about sets and weights, relying on natural notation without explicitly encoding their constraints. In the context of formal verification, however, these assumptions must be carefully reconstructed to support mechanized reasoning.

In the standard mathematical formulation of the Isolation Lemma, we consider a universe of elements U. A weight function, w, assigns a numerical value to each element in U. Typically, these weights are integers chosen uniformly and randomly from a range $\{1,\ldots,N\}$. The weight of a subset $S \subseteq U$ is then defined as the sum of the weights of its constituent elements:

$$w(S) = \sum_{x \in S} w(x) \tag{3.1}$$

To formalize this in Isabelle/HOL, we avoid defining an explicit "Universe" type. Instead, we represent the universe implicitly via type variables $'a$. More precisely, the universe is modeled as an explicit set $U ::' a$ `set`. The type variable $'a$ serves only as a carrier type, while all semantic statements about weights are restricted to elements $x \in U$. Although weight functions are total by definition in Isabelle, values outside the universe do not play any role in the proof. We introduce a type synonym to model the weight assignment. This approach enhances the readability of the subsequent definitions, and ensures type safety within the proof, which is necessary for the proof assistant.

```
type_synonym 'a weight = "'a ⇒ nat"
  (* weight maps an element to a natural number *)

definition wt :: "'a weight ⇒ 'a set ⇒ nat"  where
  "wt w S = (SUM x ∈ S. w x)"
  (* sum of weight *)
```

Listing 3.1: Definition of Weight Type and Weight Function

As shown in Listing 3.1, we define the type `'a weight` as a function mapping an element of an arbitrary type `'a` to a natural number (`nat`). Based on this type, the total weight of a set S is defined using the `wt` function.

The choice of a total function is a technical requirement of Isabelle/HOL, where all functions must be defined on the entire carrier type. Semantically, all arguments and constructions in the sequel will only depend on the values of w restricted to the universe

U. This separation between the carrier type and the explicit universe avoids treating the weight function as a partial function, which is not natively supported in Isabelle/HOL.

This modeling step illustrates a typical transition from pen-and-paper reasoning to formal verification. While informal proofs treat the definition of weights and subset sums as self-evident, formalization forces us to separate type-level carriers from explicitly quantified finite sets, and to express totality and domain restrictions precisely. These distinctions are not only technical, but crucial in enabling mechanized proof construction within a system like Isabelle/HOL.

DESIGN DECISION: NATURAL NUMBERS VS. INTEGERS

A critical modeling decision was the choice of the codomain for the weight function. While the original lemma is often stated using integers, we selected Isabelle's natural number type (`nat`). This decision is motivated by two factors:

- **Non-negativity:** In the context of the Isolation Lemma, weights are strictly positive (from $\{1,\ldots,N\}$). Using **nat** naturally enforces this constraint without requiring additional invariants to check for negative values.
- **Inductive Reasoning:** The `nat` type in Isabelle is powerful in induction heuristics which facilitate the automation of arithmetic proofs.

HANDLING FINITENESS

The definition of `wt` uses Isabelle's `sum` operator (represented as $\sum$ in the math context). It is crucial to note that in Isabelle/HOL, the summation operator is only mathematically meaningful when the set S is finite. If S were infinite, the summation would return 0 by definition in the standard library. This finiteness condition is usually implicit in informal proofs, but must be enforced explicitly in Isabelle/HOL to avoid unintended semantics.

Consequently, throughout our formalization, we do not rely on an implicit assumption that the universe is finite, which is often overlooked in pen-and-paper proofs. As it is a mandatory step in formal verification, we explicitly enforce the assumption `finite`

`S` or `finite U` locally in all lemmas involving weight calculations. Accordingly, all lemmas involving weight computations in this development explicitly assume `finite U` or `finite S`, making finiteness a first-class requirement rather than an implicit convention.

3.2 Defining Minimizers and Uniqueness

A key conceptual step in understanding the Isolation Lemma is identifying what it means for a set to be a ”minimizer”. Pen-and-paper arguments often describe this idea informally – noting that some set achieves the lowest weight – but leave the structural details implicit. To move toward a formal proof, we must refine this informal notion into a logically precise predicate.

Formally, for a family of subsets $\mathscr{F}$ and a weight function w, a set A is a minimizer if:

$$A \in \mathscr{F} \quad \text{and} \quad \forall B \in \mathscr{F},\ w(A) \leq w(B) \tag{3.2}$$

To encode this concept formally, we define a predicate `minimizer F w A` that captures both membership in $\mathscr{F}$ and optimality under the weight function. This predicate holds true if and only if the set A belongs to the family F and its weight is less than or equal to the weight of any other set B in F.

```
definition minimizer :: "'a set set ⇒ 'a weight ⇒'a set
    ⇒ bool" where
  "minimizer F w A =
    (A ∈ F ∧ (∀ B ∈ F. wt w A ≤ wt w B))"
  (* A is one minimizer of F under weight w *)

definition has_unique_minimizer :: "'a set set ⇒ 'a weight
    ⇒ bool" where
  "has_unique_minimizer F w =
    (∃! A ∈ F. minimizer F w A)"
  (* then A is the unique minimizer of F under weight w *)
```

Listing 3.2: Formal Definition of a Minimizer and Uniqueness

Listing 3.2 shows how the intuitive minimizer definition is directly translated into a formal predicate suitable for mechanized reasoning. It is important to note that this definition is purely predicate-based. The predicate `minimizer F w A` does not, by itself, assert the existence or uniqueness of such a set. These properties must be established separately under additional assumptions on the family F and the weight function w. In the setting of formal proof, such distinctions are necessary to avoid implicitly assuming the existence or uniqueness of minimizers.

The Isolation Lemma goes beyond existence – its core claim concerns the **uniqueness** of the minimizer under random weights. To formalize this probabilistic uniqueness claim, we define a predicate `has_unique_minimizer`. Here, we utilize Isabelle's unique existential quantifier $\exists!$. The expression $\exists! A.P(A)$ asserts that there exists *exactly one* element A such that the predicate $P(A)$ is true. This bridges the informal idea of uniqueness with a machine-verifiable logical form.

Thus, `has_unique_minimizer F w` states that there is exactly one set A in the family F that satisfies the minimizer condition. This concise definition serves as the target statement for our main probability bound proof in the subsequent chapters.

A USEFUL REFORMULATION OF NON-UNIQUENESS

In pen-and-paper proofs, non-uniqueness is often described informally, but in the formal setting, it is more convenient to reason with explicit witnesses. We therefore reformulate non-uniqueness using a standard logical equivalence: there is no unique minimizer if and only if two distinct sets $A \neq B$ in $\mathscr{F}$ both satisfy the `minimizer` predicate. This reformulation enables case-based reasoning in the formal proof, where fixed witnesses simplify the structural analysis of minimizers.

3.3 THE ALPHA CONSTRUCTION

The crux of the Isolation Lemma relies on a specific value associated with each element x, commonly denoted as $\alpha(x)$. Originally introduced by Mulmuley et al., this quantity provides an intuitive threshold that governs whether x contributes to a minimum-weight set. From an informal perspective, $\alpha(x)$ reflects how sensitive the total weight is to the choice of $w(x)$. Our goal in this section is to move from this intuition toward a formal expression that supports explicit reasoning in the proof assistant.

The formal definition of $\alpha(x)$ is structured to capture the marginal role of the element x within the set system $\mathscr{F}$. By partitioning the family into subsets that contain x and those that do not, the two minimization terms separately encode the weight thresholds for these two disjoint cases. This construction is designed so that the role of x can be analyzed through sets that either exclude x or remove it before evaluating the residual weight. Mathematically, $\alpha(x)$ is defined as the difference between two minimum values:

$$\alpha(x) = \min_{S \in \mathscr{F} \,:\, x \notin S} w(S) \;-\; \min_{S \in \mathscr{F} \,:\, x \in S} w(S \setminus \{x\}) \tag{3.3}$$

Intuitively, this measures the ”gap” or threshold for $w(x)$. The first term represents the minimum weight achievable without using x. The second term represents the minimum ”residual” weight required to complete a set that must contain x. Taken together, these two terms describe the boundary condition under which x determines the overall minimizer.

Formal Definition

In Isabelle/HOL, we formalize this using the LEAST operator. The two minimum values are computed independently, and their difference is taken to obtain $\alpha(x)$.

```
definition alpha :: "('a ⇒ nat) ⇒ 'a set set ⇒ 'a ⇒ nat
   " where
  "alpha w F x =
     (LEAST n. ∃ S ∈ F. x ∉ S ∧ wt w S = n)
   - (LEAST n. ∃ S ∈ F. x ∈ S ∧ wt w (S - {x}) = n)"
```

Listing 3.3: Definition of Alpha

Well-Definedness and Intended Use

In the transition from intuitive definitions to formal ones, a key difficulty is ensuring that expressions like $\alpha(x)$ are well-defined. The use of the LEAST operator in Isabelle/HOL requires explicit witnesses for the corresponding minimum conditions. This is in contrast with pen-and-paper reasoning, where such existence is often assumed without justification. In our development, such existence properties are not assumed implicitly, but are discharged later as explicit proof obligations, under suitable conditions on the family $\mathscr{F}$, in particular the existence of minimizers.

At the modeling stage, $\alpha(x)$ is treated as an abstract quantity defined in terms of minimal weights. No assumptions are made at this point about which sets realize these minimum values.

In the subsequent proof, we will show that once minimizers are known to exist, the LEAST expressions in the definition of $\alpha(x)$ can be simplified to concrete weight values. This separation reflects a standard discipline in formal verification, where definitions are kept independent of their required assumptions.

3.4 Probability Space Construction

We now formalize the last component required to move from intuitive probabilistic assumptions (PP) to a machine-verifiable framework (FP): the construction of the probability space. Mathematically, the Isolation Lemma assumes that the weight $w(x)$ for each element $x \in U$ is chosen independently and uniformly at random from the set $\{1,\ldots,N\}$. We assume $N \geq 1$, ensuring that the set $\{1,\ldots,N\}$ is non-empty and the uniform distribution is well-defined.

Modeling: Implicit and Explicit Probability

This stage illustrates a particularly significant shift from informal reasoning to explicit construction. In standard pen-and-paper proofs, probability is often treated implicitly. We simply state "Let w be a random weight function" and proceed to calculate probabilities of events. However, in a formal proof assistant like Isabelle/HOL, there is no ambiguity allowed. We cannot merely assume randomness; we must explicitly construct the underlying probability distribution (as a PMF object).

This introduces a gap between intuition and formalization. We must define a concrete probability mass function (PMF) over the function space $U \to \{1\ldots.N\}$. To bridge this gap, we make use of the `HOL-Probability` library [5], which formalizes discrete distributions using measure theory. This library makes the probability assumptions of the Isolation Lemma explicit objects of reasoning, rather than implicit conditions.

Formalizing the Product Space

To transform the intuitive statement "weights are chosen independently and uniformly" into a formal object, we construct the joint distribution as a product of marginal distributions:

1. **Uniform Component:** A uniform distribution over the finite set $\{1,\ldots,N\}$.
2. **Product Construction:** A product PMF over the finite domain U.

In pen-and-paper reasoning, such product distributions are usually left implicit. In our

formal setting, both the uniform marginals and their combination must be constructed explicitly.

The uniform marginal distribution on $\{1,\ldots,N\}$ is implemented using `pmf_of_set` [5], which constructs a uniform PMF over any finite non-empty set.

These marginals are then combined using the `Pi_pmf` operator, which creates a product distribution over the domain U, with independent sampling at each point.

```
definition w_pmf :: "nat ⇒ 'a set ⇒ 'a weight pmf" where
  "w_pmf N U =
    Pi_pmf U (0::nat) (λ _. pmf_of_set {1..N})"
  (* The probability mass function for random weights *)
```

Listing 3.4: Construction of the Weight Probability Space

As shown in Listing 3.4, `w_pmf N U` defines the probability measure over total functions of type `'a => nat`, where the random choices are made independently on U. The construction uses the `Pi_pmf` operator (mathematically Π) to combine individual uniform distributions into a single product measure. This explicitly captures the **independence** assumption, ensuring that the choice of weight for any $x \in U$ does not influence others.

A technical nuance in this definition is the argument `(0::nat)` passed to `Pi_pmf`. Since functions in Isabelle/HOL are total, they must return a value even for inputs outside their logical domain ($x \notin U$). Here, `0` serves as the default value for such out-of-domain queries. While this implementation detail is necessary to satisfy Isabelle's type system, it does not affect the mathematical validity of the lemma, as our reasoning is strictly confined to the elements within U.

By modeling the randomness as a PMF object, probabilistic events become first-class objects in the logic, allowing standard measure-theoretic reasoning, such as union bounds, to be carried out within Isabelle/HOL.

From Modeling to Proof. With the modeling phase complete, we now have a precise formal infrastructure: weight assignments over finite universes, a definition of

minimizers and uniqueness, and an explicitly constructed product probability space.

This framework resolves the ambiguities present in informal (pen-and-paper) reasoning, where independence, randomness, and uniqueness are often stated but not justified.

In the next chapter, we transition from formal modeling to fully formal verification. Each proof step will rely on the definitions developed here and will make all logical dependencies explicit, thus completing the transition from intuitive understanding to machine-checked formal proof.

4 PROOF CONSTRUCTION

In this chapter, we construct a machine-checked proof of the Isolation Lemma in Isabelle/HOL. To present the development clearly, we organize it in three layers (PP/SP/FP), bridging informal intuition and formal verification. Unlike pen-and-paper proofs, formal verification requires every mathematical step to be made fully explicit. This includes not only the main proof steps, but also background obligations such as existential witnesses, measurability, independence, type correctness, and totality.

In practice, however, a pen-and-paper (PP) proof is not translated directly into formal code. Instead, we introduce an intermediate stage, which we refer to as the structured proof (SP) level. This stage presents a structured decomposition of the argument, designed to align with Isabelle/HOL proof obligations while remaining human-readable.

Thus, each proof component in this chapter is presented in three layers:

- **PP:** The intuitive pen-and-paper sketches from the literature.
- **SP:** A structured and detailed decomposition refined to match the proof obligations of Isabelle/HOL.
- **FP:** The corresponding formalization code in Isabelle/HOL.

We organize the proof as a sequence of structural reductions. We begin by establishing the existence of minimizers and by reformulating non-uniqueness in a constructive form. Then we analyze the auxiliary quantity $\alpha(x)$, which plays a central role in isolating the dependence on individual weights. This leads to a reduction of the uniqueness failure to an explicit collision event of the form $w(x) = \alpha(x)$. Finally, we bound the probability of such events using the product probability space constructed in Chapter 3.

4.1 EXISTENCE OF MINIMIZERS

Before analyzing the uniqueness of minimizers, we must first ensure that the notion of a minimizer is well-defined. In particular, for a given family of sets $\mathscr{F}$ and a weight

function w, it is necessary to establish that at least one minimizer exists whenever $\mathscr{F}$ is non-empty.

PP. In informal proofs, this step is usually omitted. Since weights are natural numbers and the family $\mathscr{F}$ is assumed to be non-empty, the existence of a minimizer is considered self-evident.

SP. In preparation for formalization, however, we must isolate this assumption and state it precisely, since later constructs (like $\alpha(x)$ or uniqueness predicates) depend on the existence of at least one such set.

Lemma (Existence of a Minimizer). *If $\mathscr{F} \neq \emptyset$, then there exists a set $A \in \mathscr{F}$ such that A is a minimizer of $\mathscr{F}$ under the weight function w.*

FP. This lemma is formalized as:

```
lemma exist_minimizer:
  assumes "F ≠ {}"
  obtains A where "A ∈ F" "minimizer F w A"
```

Listing 4.1: Existence of Minimizer

The proof proceeds by constructing the minimal weight value using Isabelle's `LEAST` operator. First, we extract a witness $S_0 \in \mathscr{F}$ to ensure the minimum is well-defined. We then define the minimal weight m as the least n such that some $S \in \mathscr{F}$ satisfies $wt\ w\ S = n$. A corresponding set A is then chosen to realize this minimum. Finally, we show that this A has weight at most that any other set in $\mathscr{F}$, thus satisfying the `minimizer` predicate introduced in Chapter 3.

This lemma plays a foundational role: it ensures that future uses of the `LEAST` operator or comparisons among minimizers are grounded in an actual, verifiable witness. By discharging this assumption with a lemma, subsequent lemmas can avoid repeatedly re-establishing this basic existence condition.

4.2 REFORMULATING THE NON-UNIQUENESS

The Isolation Lemma asserts that with high probability, the weight function w yields a unique minimizer in $\mathscr{F}$. To prove this in a formal setting, we must first address the failure of uniqueness – i.e., the possibility that multiple sets in $\mathscr{F}$ achieve the minimum weight.

PP. In pen-and-paper proofs, non-uniqueness is usually described in a purely intuitive way: saying that a minimizer is "not unique" simply means that there exist at least two different sets in $\mathscr{F}$ that both attain the minimum weight. This idea is often expressed informally by statements such as "there are two distinct minimizers", without any further logical unpacking.

SP. To make this intuition compatible with formal reasoning, we refine it into a positive existential formulation. Instead of negating a uniqueness statement, we characterize failure of uniqueness by the explicit existence of two distinct witnesses that both satisfy the minimizer predicate.

Concretely, the minimizer of $\mathscr{F}$ under w is not unique if and only if there exist two distinct sets $A \neq B$ in $\mathscr{F}$ such that both are minimizers. In logical terms, this corresponds to replacing a negated unique-existence statement by an equivalent existential condition of the form

$$\exists A\, B.\, A \in \mathscr{F} \;\wedge\; B \in \mathscr{F} \;\wedge\; \text{minimizer}(\mathscr{F}, w, A) \;\wedge\; \text{minimizer}(\mathscr{F}, w, B) \;\wedge\; A \neq B.$$

This reformulation eliminates the outer negation and transforms the non-uniqueness into concrete objects that can be fixed in subsequent arguments.

Moreover, if A and B are both minimizers, then they must attain the same total weight. This follows directly from the definition of the minimizer predicate: each has weight at most that of any other set in $\mathscr{F}$, and hence in particular at most that of the other. This elementary but explicit observation will be used later when comparing the contribution of individual elements in two different minimizers.

Lemma (Non-Uniqueness via Two Minimizers). *Assume $\mathscr{F} \neq \emptyset$. Then the minimizer of $\mathscr{F}$ under w is not unique if and only if there exist two distinct sets $A, B \in \mathscr{F}$ such that both A and B are minimizers; i.e., there exist $A \neq B$ in $\mathscr{F}$ with*

$$\forall S \in \mathscr{F},\ w(A) \leq w(S) \ \wedge\ w(B) \leq w(S),$$

and in particular $w(A) = w(B)$.

FP. These facts are formalized as:

```
lemma not_unique_iff_exist_two:
  assumes "F ≠ {}"
  shows "¬ has_unique_minimizer F w ⟷
        (∃ A B. A ∈ F ∧ B ∈ F ∧
                minimizer F w A ∧
                minimizer F w B ∧
                A ≠ B)"
```

Listing 4.2: Non-Uniqueness Characterization

```
lemma minimizers_same_weight:
  assumes "minimizer F w A" "minimizer F w B"
  shows "wt w A = wt w B"
```

Listing 4.3: Minimizers Have Equal Weight

This reformulation replaces a negated uniqueness predicate by a positive existential statement that provides concrete witnesses. Together with the fact that any two minimizers must have the same total weight, it supplies the structural input needed for later arguments based on the symmetric difference $A \triangle B$ and the collision characterization developed in subsequent sections.

4.3 THE ALPHA CONSTRUCTION

A central ingredient in the proof of the Isolation Lemma is the auxiliary quantity $\alpha(x)$ associated with each element $x \in U$. Intuitively, $\alpha(x)$ isolates the dependence of the total weight of a set on the individual weight $w(x)$, allowing us to reason about the effect of modifying a single coordinate.

Formally, $\alpha(x)$ is defined as the difference between two minimal weight values determined by the family $\mathscr{F}$. The first term represents the minimum total weight among all sets in $\mathscr{F}$ that do not contain x. The second term represents the minimum residual weight required to form a set in $\mathscr{F}$ that contains x, excluding the contribution of x itself. This definition was introduced by Mulmuley et al. and plays a crucial role in isolating single-coordinate events.

In our formalization, $\alpha(x)$ is defined using Isabelle's `LEAST` operator, which selects the least natural number satisfying a given predicate. This allows the definition to remain witness-free at the modeling stage, while later lemmas show how the resulting `LEAST`-based expressions can be reduced to concrete arithmetic forms once suitable minimizers have been established.

4.3.1 SIMPLIFYING THE DEFINITION OF $\alpha(x)$

PP. In pen-and-paper proofs, the two minimization terms in the definition of $\alpha(x)$ are usually treated as concrete minimum values realized by suitable sets in $\mathscr{F}$. Once a minimizer exists, the least total weight among sets not containing x is simply the weight of a minimizer that avoids x. Likewise, the least residual weight among sets containing x is obtained by removing x from a minimizer that contains it. This allows the abstract `LEAST`-based definition of $\alpha(x)$ to be replaced by an ordinary arithmetic expression involving concrete minimizers.

SP. To make this intuition precise, we analyze the two `LEAST` expressions in the definition of $\alpha(x)$ separately.

First, suppose that B is a minimizer of $\mathscr{F}$ under the weight function w, and that $x \notin B$.

Since $B \in \mathscr{F}$ and attains the minimum total weight among all sets in $\mathscr{F}$, it satisfies the defining predicate of the first LEAST expression. Moreover, any other set $S \in \mathscr{F}$ with $x \notin S$ must have weight at least $w(B)$. Hence, the minimum total weight among all sets in $\mathscr{F}$ that do not contain x is attained by B, and the first LEAST expression can be shown to be equal to the concrete value $w(B)$.

Second, suppose that A is a minimizer of $\mathscr{F}$ and that $x \in A$. Removing x from A yields a residual set whose total weight equals $w(A) - w(x)$. Since $A \in \mathscr{F}$ and attains the minimum total weight among all sets in $\mathscr{F}$, any other set $S \in \mathscr{F}$ containing x must have residual weight at least $w(A \setminus \{x\})$. Therefore, the least value selected by the second LEAST operator coincides with the concrete value $w(A \setminus \{x\})$.

Together, these two observations show that both instances of the LEAST operator in the definition of $\alpha(x)$ can be eliminated whenever suitable minimizers are available. The resulting expression can then be used directly in subsequent arguments.

FP. These simplifications are formalized by the following auxiliary lemmas in Isabelle/HOL:

```
lemma minimizer_B_eq_least_not_in:
  assumes "minimizer F w B" "x ∉ B"
  shows "(LEAST n. ∃ S. S ∈ F ∧ x ∉ S ∧ wt w S = n)
         = wt w B"
```

Listing 4.4: First LEAST Eliminated by a Minimizer

```
lemma minimizer_A_eq_least_in_residual:
  assumes "finite U" "F ⊆ Pow U" "minimizer F w A" "x ∈ A
    "
  shows "(LEAST n. ∃ S. S ∈ F ∧ x ∈ S ∧ wt w (S - {x}) = n)
        = wt w (A - {x})"
```

Listing 4.5: Second LEAST Eliminated by a Minimizer

These lemmas connect the abstract LEAST expressions in the definition of $\alpha(x)$ with concrete weights of minimizers. In particular, they justify replacing both minimization

terms by explicit arithmetic values whenever a minimizer avoiding x or containing x is available.

As a result, the auxiliary quantity $\alpha(x)$ can be rewritten in terms of ordinary arithmetic expressions, which will be essential for proving its structural properties and for carrying out the probabilistic analysis in later sections.

4.3.2 Independence of $\alpha(x)$ from Individual Weights

PP. Intuitively, $\alpha(x)$ should not depend on the particular value of the single coordinate $w(x)$. Changing the value of $w(x)$ shifts the total weight of every set containing x by the same offset, while leaving the relative ordering of weights determined by the remaining elements unchanged. Hence the two minima defining $\alpha(x)$ – one over sets excluding x, and one over residual weights of sets including x – remain invariant under modifications of $w(x)$.

SP. We now formulate this independence property as a formal lemma, which will serve as a key technical input to the probabilistic argument in the next section.

Lemma (Independence of $\alpha(x)$). *Let w and w' be two weight functions such that $w(y) = w'(y)$ for all $y \in U \setminus \{x\}$. Then $\alpha(x)$ computed with respect to w is equal to $\alpha(x)$ computed with respect to w'.*

To prove this lemma, we compare the two definitions of $\alpha(x)$ under w and w'. The core observation is that for any set $S \subseteq U$ with $x \notin S$, the total weight $w(S)$ depends only on the values of w on $S \subseteq U \setminus \{x\}$, and therefore equals $w'(S)$. The same reasoning applies to residual sets of the form $S - \{x\}$, since $S - \{x\} \subseteq U \setminus \{x\}$.

Using this congruence of weights, we show that the predicates appearing in the two `LEAST` expressions defining $\alpha(x)$ are equivalent under w and w'.

$$\exists S \in F.\ x \notin S \wedge wt\ w\ S = n \quad \Longleftrightarrow \quad \exists S \in F.\ x \notin S \wedge wt\ w'\ S = n,$$

and similarly for the residual term

$$\exists S \in F.\, x \in S \wedge wt\ w\ (S - \{x\}) = n \quad \Longleftrightarrow \quad \exists S \in F.\, x \in S \wedge wt\ w'\ (S - \{x\}) = n.$$

Therefore, the two corresponding `LEAST` values coincide. Unfolding the definition of $\alpha(x)$ then yields the desired conclusion:

$$\alpha_w(x) = \alpha_{w'}(x).$$

FP. This lemma is formalized as `alpha_independent_of_coord` in Isabelle/HOL:

```
lemma alpha_independent_of_coord:
  assumes "finite U" "F ⊆ Pow U" "x ∈ U"
  shows "∀ w w'. (∀ y. y ∈ U - {x} ⟶ w y = w' y)
        ⟶ alpha (w :: 'a ⇒ nat) F x = alpha w' F x"
```

Listing 4.6: Independence of alpha(x) from the x-coordinate

This lemma is proved by establishing weight congruence on sets contained in $U - \{x\}$, transferring this congruence to the defining predicates of the two `LEAST` expressions in $\alpha(x)$, and finally unfolding `alpha_def`.

4.3.3 Collision characterization via $\alpha(x)$

PP. We now establish the key link between non-uniqueness and the auxiliary quantity $\alpha(x)$.

Assume that the minimizer of $\mathscr{F}$ under w is not unique. Then there exist two distinct minimizers $A, B \in \mathscr{F}$. Choose an element $x \in A \setminus B$. Since A and B are minimizers, their total weights coincide:

$$w(A) = w(B).$$

Moreover, in the definition of $\alpha(x)$, the minimum over sets not containing x is realized by B, while the minimum over residual weights of sets containing x is realized by

$A \setminus \{x\}$. Hence,

$$\alpha(x) = w(B) - w(A \setminus \{x\}) = w(A) - w(A \setminus \{x\}) = w(x).$$

Therefore, non-uniqueness implies the existence of an element $x \in U$ such that $w(x) = \alpha(x)$.

SP. The pen-and-paper argument above already matches closely the structure of the formal proof. The only additional ingredients required for formalization are the explicit side conditions ensuring that the two `LEAST` expressions in the definitions of $\alpha(x)$ are well-defined and can be eliminated.

Concretely, we assume that the universe U is finite and that the family $\mathscr{F}$ is a collection of subsets of U. Under these assumptions, the two auxiliary lemmas established in Section 4.3 apply: the first shows that the `LEAST` term over sets not containing x is realized by B whenever $x \notin B$, and the second shows that the residual `LEAST` term over sets containing x is realized by $A \setminus \{x\}$ whenever $x \in A$.

Together with the lemma stating that any two minimizers have equal total weight, these results allow the formal proof to mirror the pen-and-paper derivation step by step. No further combinatorial insight is required at this stage; the formal proof proceeds by assembling previously established lemmas and discharging the necessary finiteness and membership conditions.

FP. This reasoning is captured in Isabelle/HOL by the following lemma:

```
lemma two_minimizers_alpha_eq:
  assumes "finite U" "F ⊆ Pow U"
          "A ∈ F" "B ∈ F"
          "minimizer F w A" "minimizer F w B"
          "x ∈ A - B"
  shows   "alpha w F x = w x"
```

Listing 4.7: Collision Witnessed by a Single Coordinate

The proof unfolds the definition of $\alpha(x)$ and eliminates both `LEAST` expressions using the auxiliary lemmas from Section 4.3. The equality of weights $w(A) = w(B)$ follows from the fact that A and B are both minimizers. Finally, the standard splitting identity $w(A) = w(A \setminus \{x\}) + w(x)$ is used to derive the desired conclusion $\alpha(x) = w(x)$.

4.4 Probabilistic Analysis and Final Bound

PP. The probabilistic argument reduces failure of uniqueness to a single-coordinate collision event. By Section 4.3.3, if the minimizer is not unique, then there exists some $x \in U$ such that

$$w(x) = \alpha(x).$$

It therefore suffices to bound the probability of these events and sum over all $x \in U$.

For a fixed element x, the key observation is that $\alpha(x)$ depends only on the weights assigned to elements of $U \setminus \{x\}$, whereas $w(x)$ is chosen independently and uniformly from $\{1, \ldots, N\}$. Thus the event $w(x) = \alpha(x)$ can occur with probability at most $1/N$.

Applying the union bound over all $x \in U$ yields

$$\Pr\big[\text{the minimizer is not unique}\big] \leq \frac{|U|}{N},$$

and therefore

$$\Pr\big[\text{the minimizer is unique}\big] \geq 1 - \frac{|U|}{N}.$$

This is the essential probabilistic idea behind the Isolation Lemma.

SP. Fix $x \in U$. To bound the event $\{\, w(x) = \alpha(x) \,\}$, we first decompose it according to the possible values of the coordinate $w(x)$:

$$\{\, w(x) = \alpha(x) \,\} \;=\; \bigcup_{k=1}^{N} \{\, \alpha(x) = k \;\wedge\; w(x) = k \,\}.$$

Since the union is finite, it suffices to bound the probability of each summand.

Now fix $k \in \{1, \ldots, N\}$. By Section 4.3.2, the quantity $\alpha(x)$ depends only on the weights assigned to elements of $U \setminus \{x\}$. On the other hand, the random variable $w(x)$ depends only on the x-coordinate. Because the underlying distribution is a product probability space, these two random variables are independent. Hence

$$\Pr[\, \alpha(x) = k \wedge w(x) = k \,] = \Pr[\, \alpha(x) = k \,] \cdot \Pr[\, w(x) = k \,].$$

To justify this factorization, we verify that the relevant random variables and events are measurable. This requires showing that finite sums of coordinates are measurable, that predicates of the form $\exists S \in F.\ \cdots$ define measurable sets, and that the `LEAST`-based construction of $\alpha(x)$ preserves measurability. Once these facts are established, the standard independence lemmas of Isabelle/HOL apply.

Since $w(x)$ is uniformly distributed on $\{1, \ldots, N\}$, we have

$$\Pr[\, w(x) = k \,] = \frac{1}{N} \qquad \text{for each } k \in \{1, \ldots, N\}.$$

Therefore,

$$\Pr[\, \alpha(x) = k \wedge w(x) = k \,] = \Pr[\, \alpha(x) = k \,] \cdot \frac{1}{N}.$$

Summing over all $k \in \{1, \ldots, N\}$ and using

$$\sum_{k=1}^{N} \Pr[\, \alpha(x) = k \,] \leq 1,$$

we obtain

$$\Pr[\, w(x) = \alpha(x) \,] \leq \frac{1}{N}.$$

Finally, by Section 4.3.3,

$$\{\text{the minimizer is not unique}\} \subseteq \bigcup_{x \in U} \{\, w(x) = \alpha(x) \,\}.$$

Applying the union bound over the finite set U, we obtain

$$\Pr\big[\text{the minimizer is not unique}\big] \leq \sum_{x \in U} \Pr[w(x) = \alpha(x)] \leq \frac{|U|}{N}.$$

Equivalently,

$$\Pr\big[\text{the minimizer is unique}\big] \geq 1 - \frac{|U|}{N},$$

which is exactly the desired statement of the Isolation Lemma.

FP. In Isabelle/HOL, the above structural steps are realized by the following core lemmas and theorems:

(a) Coordinate support and point probabilities. We first show that for any $x \in U$, the random variable $w(x)$ takes values almost surely in $\{1, \ldots, N\}$. This yields the following support lemma:

```
lemma w_pmf_component_in_range:
  assumes "x ∈ U" "N ≥ 1" "finite U"
  shows "set_pmf (map_pmf (λ w. w x) (w_pmf N U)) ⊆ {1..N
     }"
```

Listing 4.8: Coordinate Support

Based on this, we derive the uniform point probability formula for each $k \in \{1, \ldots, N\}$ (appearing as a subgoal in the main proof):

```
measure_pmf.prob (w_pmf N U) {w. w x = k} = 1 / real N
```

Listing 4.9: Point Probability of One Coordinate

(b) Measurability of $\alpha(x)$. To apply probabilistic reasoning inside Isabelle/HOL, we must show that the auxiliary quantity $\alpha(x)$ is measurable. In informal presentations,

this point is usually left implicit, but in the formal development it must be established explicitly, since $\alpha(x)$ is defined through a LEAST-based construction. The argument proceeds by characterizing the relevant level sets using finite combinations of measurable predicates over the family $\mathscr{F}$, thereby placing the subsequent probability bounds on a rigorous measure-theoretic footing.

This measurability is captured by:

```
lemma alpha_measurable:
  assumes "finite U" "F ⊆ Pow U" "x ∈ U"
  shows "(λ f. alpha (λ y. if y ∈ U-{x} then f y else 0) F
     x)
         ∈ (Π_M (U-{x}) (λ _. count_space UNIV)) →_M (
            count_space UNIV)"
```

Listing 4.10: Measurability of Alpha

This proof relies on measurability of finite sums, finite existential predicates, and the measurability theorem for the `LEAST` operator.

(c) Independence lemma. Next, we lift coordinate-wise independence to the level of random variables by showing that $w(x)$ is independent of $\alpha(x)$, since the latter depends only on the remaining coordinates:

```
prob_space.indep_var (measure_pmf (w_pmf N U))
  (count_space UNIV) (λ w. w x)
  (count_space UNIV) (λ w. alpha w F x)
```

Listing 4.11: Independence of Coordinate and Alpha

This step combines restriction lemmas for product measures with the previously established lemma `alpha_independent_of_coord`, ensuring that $\alpha(x)$ depends solely on the weights assigned to $U - \{x\}$.

(d) Main theorem: Isolation Lemma. Finally, combining the single-coordinate probability bound with the union bound yields the formalized version of the Isolation Lemma:

```
theorem isolation_lemma_main:
  fixes U :: "'a set" and F :: "'a set set" and N :: nat
  assumes "finite U" "F ⊆ Pow U" "F ¬eq {}" "N ≥ 1"
  shows "measure_pmf.prob (w_pmf N U) {w.
     has_unique_minimizer F w}
         ≥ 1 - real(card U) / real N"
```

Listing 4.12: Main Theorem: Isolation Lemma

Taken together, these lemmas provide the formal probabilistic infrastructure needed for the final argument. With measurability and independence established explicitly in Isabelle/HOL, the Isolation Lemma follows by combining the single-coordinate bound with the union bound.

4.5 Chapter Summary

In this chapter, we presented a fully formal proof of the Isolation Lemma in Isabelle/HOL, by systematically refining a classical proof into structured proof steps and finally machine-checked formal Isabelle/HOL code. We reformulated non-uniqueness as the existence of two distinct minimizers, introduced the auxiliary quantity $\alpha(x)$, and showed how it captures the precise collision event responsible for ties. A key technical step was to prove that $\alpha(x)$ is independent of the individual weight $w(x)$, allowing the argument to reduce to single-coordinate probabilistic events. Combining this with the probabilistic machinery developed over the product probability space, we derived the standard lower bound

$$1 - \frac{|U|}{N}$$

for the probability that the minimizer is unique.

5 Perfect Matching Problem

5.1 Introduction to Randomized Parallel Matching

We now turn to the main application of the formalized Isolation Lemma: a machine-checked correctness argument for a randomized parallel algorithm for the Perfect Matching problem in bipartite graphs.

The classical algorithmic background of Perfect Matching and the role of isolation were reviewed in Chapter 2. The focus of the present chapter is therefore not to repeat that background, but to show how the formalized Isolation Lemma fits into the machine-checked correctness argument.

More specifically, the chapter proceeds in two stages. First, we establish a conditional correctness argument: assuming that random weighting isolates a unique minimum-weight perfect matching, we show that this matching can be recovered through algebraic tests based on determinants, multiplicities, and minors. Second, we instantiate the formalized Isolation Lemma to the family of perfect matchings, thereby proving that this uniqueness condition holds with high probability under random edge weights. Together, these two parts yield the probabilistic correctness argument for the randomized parallel algorithm.

Decision versus Construction. For the Perfect Matching problem, there is an important distinction between the *decision* version, which asks whether a perfect matching exists, and the *construction* version, which asks how to recover the edges of such a matching. The decision version admits efficient randomized parallel treatment through algebraic methods based on matrix determinants.

By contrast, the construction version is more difficult in a parallel setting, since identifying the matching requires globally consistent choices rather than merely certifying existence. This difficulty is amplified by the presence of many symmetric perfect matchings. Without a mechanism for selecting a single distinguished solution, parallel local decisions may fail to assemble into one global matching.

5.2 Algebraic Modeling via the Tutte/Edmonds Matrix

To formalize the matching application, we first translate the combinatorial search for perfect matchings into an algebraic problem amenable to parallel reasoning. The standard approach utilizes the *Tutte matrix* [2] (often referred to as the Edmonds matrix in the bipartite case) to encode the graph into matrix form. In the present development, this encoding is used not only to detect perfect matchings, but also to represent their total weights algebraically, so that the minimum-weight matching can later be recovered from the determinant-based information.

5.2.1 Matrix Construction in Isabelle/HOL

In our formalization, we consider a bipartite graph $G = (U \cup V, E)$ with $|U| = |V| = n$. We write the weight of an edge (i, j) as $w_{i,j}$. In the `bipartite_graph` locale, the matrix D is constructed such that each entry $D_{i,j}$ encodes both the existence of an edge and its weight:

```
definition D_entry ::
    "nat ⇒ nat ⇒ int" where
    "D_entry i j = (if (i,j) ∈ E then (2::int) ^ (w i j)
       else 0)"
```

Listing 5.1: Matrix Entry Definition

The choice of $2^{w_{i,j}}$ as the matrix entry, as proposed in [2], [32], is the key algebraic design decision in this construction. It ensures that the total weight of a perfect matching is reflected in the exponent of 2 appearing in the corresponding term of the determinant expansion. This algebraic encoding allows the determinant to aggregate information about all feasible matchings simultaneously through its powers-of-2 structure.

5.2.2 Determinant and Perfect Matchings

A perfect matching in a bipartite graph can be formally defined as a permutation p of the set $\{0,\ldots,n-1\}$ such that every pair $(i,p(i))$ is a valid edge in E. Our formal definition in Isabelle/HOL captures this requirement:

```
definition is_perfect_matching ::
    "(nat ⇒ nat) ⇒ bool" where
  "is_perfect_matching p ⟷
    (p permutes {0..<n} ∧ (∀ i < n. (i, p i) ∈ E))"
```

Listing 5.2: Perfect Matching Definition

The determinant of matrix D, defined via the Leibniz formula, serves as the primary algebraic object in the extraction argument:

$$\det(D) = \sum_{p \in S_n} \operatorname{sgn}(p) \prod_{i=0}^{n-1} D_{i,p(i)} \tag{5.1}$$

By substituting the definition of $D_{i,j}$, any term in the sum is non-zero if and only if the permutation p corresponds to a valid perfect matching. For such a permutation, the contribution to the determinant is:

$$\operatorname{sgn}(p) \cdot \prod_{i=0}^{n-1} 2^{w_{i,p(i)}} = \operatorname{sgn}(p) \cdot 2^{\sum w_{i,p(i)}} = \operatorname{sgn}(p) \cdot 2^{weight(p)} \tag{5.2}$$

If p does not represent a perfect matching, at least one $D_{i,p(i)}$ is 0, causing the entire product to vanish. This mapping shows that the determinant is a signed sum over all perfect matchings, with each matching contributing a term whose power of 2 records its total weight. Figure 5.1 illustrates this correspondence on a small example: the highlighted perfect matching on the graph corresponds to the highlighted entries in the matrix.

This is the key algebraic encoding used in the next section, where we show that, under the uniqueness condition, the minimum matching weight is reflected in the 2-adic multiplicity of the determinant.

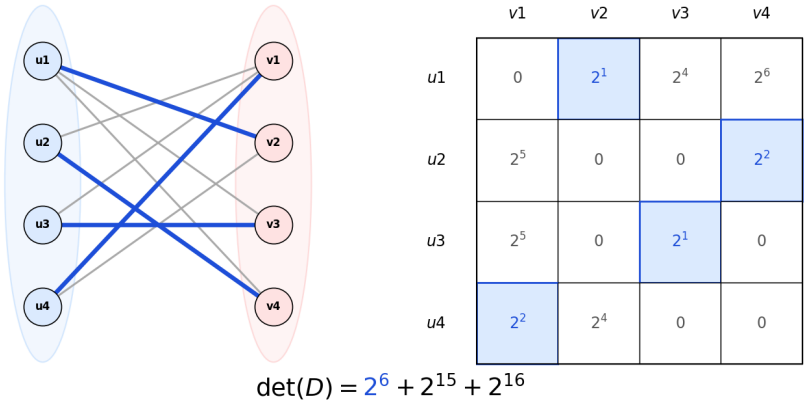

Fig. 5.1: Graph-to-matrix encoding for a weighted bipartite graph. Each nonzero matrix entry corresponds to an edge and is given by $2^{w_{i,j}}$, while absent edges contribute 0. The highlighted entries correspond to the highlighted minimum-weight perfect matching.

5.3 The Weight-to-Multiplicity Bridge

The previous section encoded the weights of perfect matchings into the determinant expansion of the Tutte matrix. We now establish the central algebraic bridge of the chapter: under the uniqueness condition, the minimum perfect-matching weight can be recovered from the 2-adic multiplicity of the determinant. This global invariant will later serve as the basis for the edge-wise extraction criterion.

5.3.1 The Non-Cancellation Principle

In the Leibniz expansion of $\det(D)$, each perfect matching p contributes a term of the form $\operatorname{sgn}(p) \cdot 2^{weight(p)}$. The recovery of the minimum weight W_0 from this expansion relies crucially on uniqueness. If several perfect matchings attain the same minimum weight W_0, then the coefficient of 2^{W_0} becomes a signed sum over all such matchings, and this sum may vanish by cancellation. In that case, the minimum weight can no longer be identified through the 2-adic multiplicity of the determinant. By contrast, if the minimum-weight perfect matching is unique, its contribution survives as a single non-zero term, and no term of higher weight can cancel it. This is exactly why the uniqueness guarantee provided by the Isolation Lemma is essential for the weight-to-multiplicity argument.

Let P denote the set of all perfect matchings of G. Formally, if W_0 is the unique minimum weight, $\det(D)$ can be expressed as:

$$\det(D) = \operatorname{sgn}(p_{\min})2^{W_0} + \sum_{p \in P \setminus \{p_{\min}\}} \operatorname{sgn}(p)2^{weight(p)} \tag{5.3}$$

Since for all other matchings p, $weight(p) > W_0$, every term in the summation is a multiple of 2^{W_0+1}. Therefore, the determinant is divisible by 2^{W_0} but not by 2^{W_0+1}. Equivalently, the 2-adic multiplicity of the determinant is exactly W_0. This is the algebraic fact formalized in Isabelle/HOL in the lemmas below.

5.3.2 Formalization via Multiplicity

To extract this minimum weight in Isabelle/HOL, we use the `multiplicity` function from the number theory library. For an integer x, the multiplicity of 2 in an integer x is the largest exponent m such that 2^m divides x. In the present setting, taking the multiplicity of 2 in the determinant quantity formalized as `det_D` yields exactly the minimum weight W_0.

We formalized this connection through several key lemmas. First, `multiplicity_2_abs` establishes the base case for powers of 2:

```
lemma multiplicity_2_abs:
  fixes x :: int
  assumes "abs x = 2 ^ k"
  shows "multiplicity 2 x = k"
```

Listing 5.3: Multiplicity of Power of 2

We then establish the main theorem relating the determinant to the unique minimum weight:

```
lemma unique_det_power_of_weight:
  assumes "unique_min_weight_condition"
  shows "multiplicity 2 det_D = min_weight_val"
```

Listing 5.4: Multiplicity of Determinant

This result is proven in Isabelle by applying `multiplicity_eqI`, which requires showing that 2^{W_0} divides the sum and 2^{W_0+1} does not. The proof handles the summation by splitting it into the unique minimum term and the remaining higher-weight terms, formalizing the intuitive non-cancellation argument presented earlier. This theorem provides a global algebraic characterization of the minimum perfect-matching weight. The next section refines this global information into a local criterion for deciding whether a given edge belongs to the unique minimum perfect matching.

5.4 Parallel Edge Extraction Logic

The previous section provided a global algebraic characterization of the minimum perfect-matching weight. We now refine this information into an edge-wise extraction criterion. The final stage of the extraction argument is to determine which edges $(i, j) \in E$ belong to the unique minimum-weight perfect matching $p_{\min}$. Crucially, this step must remain parallelizable, so each edge must be testable independently of the others.

5.4.1 Minors and Subgraph Matchings

For any edge (i, j), we consider the minor $D_{i,j}$, which is the submatrix obtained by deleting the i-th row and j-th column of D. Let $u_i \in U$ and $v_j \in V$ denote the vertices indexed by i and j, respectively. Algebraically, the determinant of this minor encodes the perfect matchings of the reduced graph $G \setminus \{u_i, v_j\}$ in exactly the same weighted form as the full determinant encodes perfect matchings of G. If an edge (i, j) belongs to a perfect matching p in G, then $p \setminus \{(i, j)\}$ must be a perfect matching in the reduced graph.

In our formalization, we define `minor_sum i j` to represent the contribution of all matchings containing the edge (i, j). This relationship is captured by the lemma `minor_sum_decomposition`:

```
lemma minor_sum_decomposition:
  assumes "i < n" "j < n" "(i,j) ∈ E"
  shows "minor_sum i j = D_entry i j * cofactor_sum i j"
```

Listing 5.5: Minor Sum Decomposition

This decomposition isolates the contribution of the edge (i, j) from the remaining matching structure. Figure 5.2 illustrates this reduction on a small example: selecting an edge in the highlighted perfect matching corresponds to selecting its matrix entry and deleting the associated row and column.

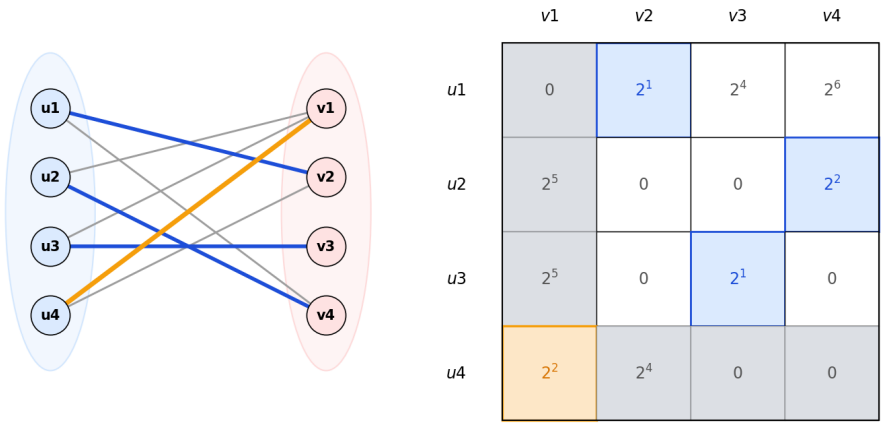

Fig. 5.2: Reduction from a selected matching edge to the corresponding matrix position. The highlighted edge in the minimum-weight perfect matching corresponds to the highlighted matrix entry, while the shaded row and column indicate the deletion used to form the associated minor.

It is precisely this separation that makes it possible to turn the global minimum-weight condition into a local test for edge membership.

5.4.2 The Parallel Decision Predicate

Under the uniqueness condition, an edge (i, j) belongs to $p_{\min}$ exactly when the minimum weight of a perfect matching in the reduced graph, together with the weight of (i, j) itself, reconstructs the global minimum weight W_0. Thus the global invariant obtained in the previous section can be refined into an edge-wise decision predicate.

This core algebraic criterion can be expressed by the identity:

$$w_{i,j} + \mathrm{mult}_2\big(\det(D_{i,j})\big) = W_0 \tag{5.4}$$

where W_0 denotes the minimum perfect-matching weight.

To connect the informal identity in (5.4) with the Isabelle statement, we use the decomposition established in Lemma 5.5:

$$\texttt{minor_sum}\ i\ j = \texttt{D_entry}\ i\ j \cdot \texttt{cofactor_sum}\ i\ j.$$

Here, `D_entry` i j is the matrix entry associated with the edge (i, j), namely $\pm 2^{w_{i,j}}$, so its 2-adic multiplicity is exactly $w_{i,j}$. Thus, whenever `cofactor_sum` $i\ j \neq 0$, additivity of multiplicity gives

$$\mathrm{mult}_2(\texttt{minor_sum}\ i\ j) = w_{i,j} + \mathrm{mult}_2(\texttt{cofactor_sum}\ i\ j).$$

Accordingly, in the formal development, the global minimum weight W_0 is represented by `min_weight_val`, while the minor contribution associated with deleting row i and column j is captured by `cofactor_sum i j`.

The informal criterion

$$w_{i,j} + \mathrm{mult}_2\big(\det(D_{i,j})\big) = W_0$$

is captured in the formal development by the following edge-wise test:

```
theorem edge_test_iff_correct:
  assumes p_min_def:
    "is_perfect_matching p_min"
    "matching_weight p_min = min_weight_val"
  assumes p_min_unique:
    "∀ q. is_perfect_matching q ∧ matching_weight q
            = min_weight_val ⟶ q = p_min"
  shows "p_min i = j ⟷
         (cofactor_sum i j ≠ 0 ∧ w i j + multiplicity 2 (
            cofactor_sum i j) = min_weight_val)"
```

Listing 5.6: Correctness of Edge Extraction

This theorem gives the central conditional correctness statement for the extraction stage of the algorithm. Under the uniqueness premise, each edge (i, j) can be tested independently through its associated minor-based algebraic condition. Hence, once the uniqueness premise is satisfied, the entire matching can be reconstructed in parallel.

What remains is to justify this premise probabilistically. The next section therefore applies the formalized Isolation Lemma to the family of perfect matchings, showing that the required uniqueness condition holds with high probability under random edge weights.

5.5 Probabilistic Correctness via Isolation Lemma

The extraction theorem of the previous section is conditional on the existence of a unique minimum-weight perfect matching. The purpose of the present section is to justify this uniqueness premise probabilistically. While Chapter 4 established the Isolation Lemma for an arbitrary finite family of sets, we now instantiate that general result to the family of perfect matchings in a bipartite graph.

5.5.1 Mapping Graphs to Sets

To apply the Isolation Lemma, we first define a formal mapping between graph-theoretic structures and the set-theoretic framework. In the abstract set-family formulation of the Isolation Lemma, the universe is identified with the edge set E, and the family $\mathscr{F}$ is defined as the collection of all edge sets that constitute a valid perfect matching:

```
definition perfect_matching_family ::
    "nat ⇒ (nat * nat) set ⇒ ((nat * nat) set) set" where
    "perfect_matching_family n E =
      {matching_to_set n p | p.
       p permutes {0..<n} ∧ (∀ i < n. (i, p i) ∈ E)}"
```

Listing 5.7: Perfect Matching Family

To transfer the Isolation Lemma to the graph setting, we must show that the weight of a matching in the graph model coincides with the weight of the corresponding set in the abstract isolation model. This is established by the lemma `weight_equivalence`. Furthermore, the lemma `matching_to_set_inj` ensures that distinct perfect matchings are represented by distinct edge sets in the set-family model.

These lemmas together provide the interface between the graph-theoretic formulation of perfect matching and the abstract set-family formulation used in Chapter 4. They ensure that the Isolation Lemma can be transferred to the perfect-matching setting without changing the underlying weight semantics.

5.5.2 Interface to the Isolation Lemma

The key interface theorem is the lemma `pm_iso_equivalence`. It identifies the graph-theoretic uniqueness condition required by the extraction argument with the abstract predicate `has_unique_minimizer` used in Chapter 4. This equivalence is what allows the general Isolation Lemma to be lifted to the specific case of bipartite perfect matchings.

```
lemma pm_iso_equivalence:
  fixes w_func :: "nat * nat ⇒ nat"
  shows "bipartite_graph.unique_min_weight_condition n (λ
     u v. w_func(u,v)) E
        ⟷ has_unique_minimizer (perfect_matching_family n
            E) w_func"
```

Listing 5.8: Equivalence between perfect-matching uniqueness and the abstract minimizer predicate

5.5.3 Final Probability Bound and End-to-End Correctness

The theorem isolation_lemma_for_perfect_matching provides the machine-checked lower bound on the success probability of the isolation step:

```
theorem isolation_lemma_for_perfect_matching:
  fixes N :: nat
  assumes "N ≥ 1"
  assumes "∃ p. is_perfect_matching p"
  assumes "E ≤ {0..<n} * {0..<n}"
  shows "measure_pmf.prob (w_pmf N E)
      {w. unique_min_weight_condition n (λ u v. w (u,v)) E
         }
      ≥ 1 - (real (card E) / real N)"
```

Listing 5.9: Isolation Lemma instantiated for Perfect Matching

This theorem shows that, for a graph with edge set E and edge weights chosen uniformly from $\{1,\ldots,N\}$, the event that a unique minimum-weight perfect matching exists has probability at least $1-\frac{|E|}{N}$.

However, this isolation statement alone is not yet the final algorithmic correctness result. The extraction theorem from Section 5.4 shows only that *if* a unique minimum-weight perfect matching exists, then the minor-based edge test reconstructs it correctly.

To obtain the randomized correctness statement for the algorithm, we must combine these two results into a single end-to-end theorem.

We therefore prove the following final correctness theorem:

```
theorem edge_test_recovers_with_high_probability:
  fixes N :: nat
  assumes N_ge_1: "N ≥ 1"
  assumes PM_exists: "∃ p. bipartite_graph.
     is_perfect_matching n E p"
  assumes E_bounds: "E ≤ {0..<n} * {0..<n}"
  shows
    "measure_pmf.prob (w_pmf N E)
      {w. edge_test_recovers_matching n E w}
     ≥ 1 - real (card E) / real N"
```

Listing 5.10: End-to-end high-probability correctness of edge extraction

The proof of this final theorem follows the logical structure of the development in the previous two sections. Section 5.4 established a conditional correctness result: whenever the random weight assignment produces a unique minimum-weight perfect matching, the minor-based edge test recovers exactly that matching. Section 5.5 then showed, via the Isolation Lemma, that this uniqueness event occurs with probability at least $1 - \frac{|E|}{N}$.

The theorem above combines these two ingredients into a single probabilistic correctness statement for the randomized parallel matching procedure. That is, the algorithm is correct for every weight assignment satisfying the uniqueness condition, and the Isolation Lemma guarantees that such weight assignments occur with probability at least $1 - \frac{|E|}{N}$. This yields a Monte Carlo-style correctness bound: the algorithm succeeds with probability at least $1 - \frac{|E|}{N}$, and fails with probability at most $\frac{|E|}{N}$. Hence the connection between the isolation argument and the extraction argument is not merely informal; it is captured by a final machine-checked theorem.

5.6 CHAPTER SUMMARY

In this chapter, we developed the perfect-matching application of the formalized Isolation Lemma in Isabelle/HOL. The argument proceeds in two stages.

First, we established a conditional correctness argument. By encoding edge weights as powers of 2 in the Tutte/Edmonds matrix, we showed that the determinant captures the weighted structure of all perfect matchings. We then proved the weight-to-multiplicity bridge, showing that under the uniqueness condition, the minimum matching weight can be recovered from the 2-adic multiplicity of the determinant. Building on this global invariant, we formalized a minor-based edge-extraction criterion, which yields a parallel test for deciding whether a given edge belongs to the unique minimum-weight perfect matching.

Second, we established the probabilistic justification of the uniqueness premise required by this extraction argument. By mapping perfect matchings to the abstract set-family framework of Chapter 4, we instantiated the general Isolation Lemma to the family of perfect matchings in a bipartite graph. This yielded a machine-checked lower bound showing that a unique minimum-weight perfect matching exists with probability at least $1 - \frac{|E|}{N}$ under random edge weights.

Combining these two parts, we obtained the final probabilistic correctness argument for the randomized parallel algorithm. With high probability, random weighting isolates a unique minimum-weight perfect matching. Under this condition, the minor-based algebraic test reconstructs exactly that matching in parallel. Thus this chapter completes the transition from the abstract probabilistic formalization of Chapter 4 to a concrete machine-checked correctness argument for randomized parallel matching.

6 Conclusion and Future Work

Having completed both the formal proof of the Isolation Lemma and its application to a machine-checked correctness argument for the Perfect Matching problem, we now turn to the broader contributions, methodological lessons and reflections, and future directions of the report.

6.1 Summary of Research Achievements

This report has presented a machine-checked formalization of the Isolation Lemma in Isabelle/HOL, together with its application within a two-stage machine-checked correctness argument for a randomized parallel algorithm for the Perfect Matching problem. In doing so, we have established a rigorous foundation for the formalization and verification of randomized algorithmic arguments.

The primary contributions of this work are:

- **Rigorous Foundation for Isolation**: We constructed the relevant product probability space and proved the $1 - |U|/N$ success bound, making all assumptions about measurability and independence explicit.
- **Algorithmic Correctness**: We established a machine-checked conditional correctness argument for extracting the unique minimum-weight perfect matching via the Tutte matrix, multiplicity reasoning, and minors, and then combined it with the formalized Isolation Lemma to obtain the final probabilistic correctness argument.
- **Bridging the Formalization Gap**: We employed a layered PP/SP/FP approach to connect informal folklore-style proofs with fully formal machine-checked developments.

6.2 Methodological Lessons from the Formalization

The formalization of the Isolation Lemma and its application to the Perfect Matching problem yielded several methodological lessons about the verification of randomized algorithms in Isabelle/HOL.

6.2.1 Making Randomness Explicit in Isabelle/HOL

A primary methodological lesson from this development is that randomness could not be treated as an implicit background assumption. Instead, the random weight assignment itself had to be represented explicitly within the formal system. In particular, in standard presentations of the Isolation Lemma, the independence of weight assignments and the structure of the underlying probability space are typically assumed without explicit construction. In our case, this meant that the distribution of edge or element weights, the associated product probability space, and the event of having a unique minimum-weight solution all had to be specified formally inside the development, since none of these could be treated implicitly without breaking the chain of machine-checked reasoning.

Using the `Pi_pmf` framework, we modeled independent random weight assignments in a form that supported later reasoning about independence, measurability, and probability bounds. The key lesson is that in a proof assistant, this probabilistic infrastructure must be constructed explicitly and early, since later arguments (such as uniqueness and probability bounds) depend critically on these definitions being available in a machine-checkable form.

6.2.2 The Role of the Structured Proof Layer

A second methodological lesson was that the **Structured Proof (SP)** layer served as a useful bridge between informal intuition and fully formal proof development. While SP is not itself a logical innovation, it acts as a diagnostic tool that decomposes compressed folklore-style arguments into explicit intermediate steps. This

methodology revealed that many steps treated as obvious in the literature require non-trivial formal justification, particularly the measurability of the threshold $\alpha(x)$, which is typically left implicit in pen-and-paper presentations.

6.2.3 Consistency of Representations

In Chapter 5, we encountered a representation gap between graph-theoretic permutations and set-theoretic edge families. Formalizing this connection as an injective mapping was a critical part of the development. It showed that the correctness of an algorithm depends not only on the logical steps of the argument, but also on the consistency of data representations used across different formal settings. Without a formal proof of injectivity, the probabilistic guarantees could not be transferred to the bipartite matching model, effectively blocking the correctness argument.

6.2.4 Type Safety and Measurability Obligations

A final methodological lesson concerns the additional burden of rigor imposed by type safety and measurability obligations. In particular, reconciling the types `nat` and `int`, while also discharging explicit measurability proofs, added substantial formal overhead to the development. While explicit coercions, type conversions, and measurability obligations increased the proof burden, they also eliminated classes of subtle errors that would have remained hidden in an informal proof. This illustrates that seemingly low-level issues, such as type coercions and measurability conditions, can become structurally necessary components of the proof rather than incidental technicalities.

Summary. Taken together, these observations show that the formalization of the Isolation Lemma is not merely a transition of an existing argument into Isabelle/HOL, but a refinement of its logical structure. Several aspects of the argument that appear straightforward in informal presentations—such as independence assumptions, measurability conditions, and representation consistency—become explicit proof obligations in the formal setting. This suggests that formal verification not only

increases rigor, but can also clarify the underlying structure of randomized algorithms by exposing hidden dependencies and implicit assumptions.

6.3 Reflections on Formal Methods in an Independent Research Context

This project required substantial engagement with material that does not usually form part of the standard curriculum in theoretical computer science. In particular, the use of Isabelle/HOL, the conventions of AFP-style formalization, and the broader perspective of formal mathematics were not simply auxiliary tools added at the end of the project, but became part of the research process itself.

As a result, part of this work necessarily took the form of independent study conducted alongside the formal development. This aspect of the project was significant not because it introduced unfamiliar terminology alone, but because it required learning how formal methods organize mathematical reasoning at a level of precision not normally demanded in standard coursework.

6.3.1 Formal Methods beyond the Standard Curriculum

One important reflection arising from this process is that formal methods are best understood not merely as a technical language, but as a different mode of engaging with mathematical arguments. At the beginning of the project, it was natural to view Isabelle/HOL primarily as a system for encoding already understood proofs. However, working through the formalization made it clear that the proof assistant does not simply record reasoning after the fact.

Instead, it forces a re-examination of definitions, assumptions, and proof dependencies at every stage. In this respect, learning Isabelle/HOL and formal mathematics in an independent research context was not separate from the research itself, but became one of the means through which the underlying mathematics was more fully understood. This shift suggests that engaging with a proof assistant does not merely formalize understanding, but actively reshapes what it means to

understand a mathematical argument.

6.3.2 From Following Proofs to Reconstructing Proofs

A second reflection concerns the distinction between following an informal proof and reconstructing it in a formal setting. Before undertaking the formalization, it was easy to regard a proof as understood once its main argument and overall intuition were clear. The experience of formalization showed that this standard is often insufficient for establishing full correctness.

Many steps that appear routine in pen-and-paper presentations depend on side conditions, representational choices, or intermediate facts that remain unstated unless one is forced to supply them explicitly. This was particularly evident in the treatment of probabilistic constructions, measurability conditions, and the transfer between different representations of the same mathematical object.

The project therefore sharpened the distinction between intuitive understanding and fully articulated understanding: to formalize a proof is not simply to restate it more rigorously, but to reconstruct the structure that makes each step valid. More importantly, it suggests that formalization can play a role in the development of theoretical results themselves, by making hidden assumptions visible and forcing a more robust formulation of arguments.

6.3.3 Implications for Future Theoretical Research

This project also has broader implications for future work in theoretical computer science and proof-oriented research. It has encouraged a more careful attitude toward definitions, hidden assumptions, and interfaces between different mathematical representations. More generally, it suggests that formalization can serve not only as a tool for verification, but also as a form of research training.

By requiring explicit control over proof structure, formal methods cultivate habits of precision that remain valuable even outside a proof assistant. In that sense, the project has contributed not only a formal development in Isabelle/HOL, but also a deeper appreciation of how rigorous theoretical reasoning is constructed, communicated, and

checked.

6.4 Future Work and AFP Submission

The results of this project are currently being prepared for submission to the **Archive of Formal Proofs (AFP)**. As a curated collection of Isabelle/HOL developments, the AFP would provide a natural venue for turning this formalization into a reusable entry that future work can build on. In particular, an AFP submission would make the Isolation Lemma available as a reusable component for later formalizations of randomized algorithms and related probabilistic arguments.

Future research could extend this work in several directions:

- **Further Isolation-Based Applications:** A natural next step is to extend the present development to other algorithmic settings that follow the same proof pattern, namely using random weighting to isolate a unique combinatorial object from a larger family of candidates. Besides bipartite perfect matching, promising examples include isolation-based counting and witness extraction for subset families, where uniqueness is used to reduce a large solution space to a single distinguished object with high probability [2], [17].
- **Application to Other Combinatorial Structures**: Beyond bipartite perfect matchings, it would be valuable to investigate how readily the current Isabelle/HOL formalization transfers to other combinatorial families in which weighted subsets are used to distinguish a unique optimum. In particular, this concerns the reuse of the representation of weighted subsets, the probabilistic framework based on `Pi_pmf`, and the uniqueness argument. Possible candidates include spanning-tree-type families or other subset-based structures, where the main question is not merely whether isolation holds abstractly, but how much of the present Isabelle/HOL infrastructure can be reused with only limited redesign.
- **Proof Automation for Probabilistic Reasoning**: Future efforts could also investigate better automation support for recurring proof obligations

encountered in this development, such as measurability conditions, summation manipulations, and routine divisibility reasoning. This would reduce the manual effort required to formalize probabilistic proofs of the kind studied in this report.

Overall, the present development suggests that formal verification of randomized algorithm theory can proceed beyond isolated examples toward reusable formal infrastructure. In that sense, the Isolation Lemma is not only a case study, but also a starting point for broader formal developments in randomized and probabilistic reasoning.

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